Intro Machine Learning

~3 contact hours

CHAPTERS

- Machine Learning Applications
- Learning Paradigms in ML
- Metrics

WHY THIS MODULE MATTERS

Provides a general introduction to machine learning and main evalutation metrics

Machine Learning

"An important feature of a **learning machine** is that its teacher will often be very largely ignorant of quite what is going on inside […]. The learning process may be regarded as a search for a form of behaviour which will satisfy the teacher (or some other criterion)."

A. Turing

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E."

T. Mitchell

Machine Learning

Machine Learning is currently one of the **most dynamic and transformative areas** within Artificial Intelligence and Computer Science.

Leading global ICT companies—including **Google, Meta (Facebook), IBM, Baidu, Amazon, Microsoft, and Disney**—are heavily investing in Machine Learning technologies.

Why ?

Machine Learning

Because these companies generate **massive volumes of data**—and Machine Learning provides the tools to **extract value, automate decisions, and create intelligent services**.

Machine Learning: Spam Detection

Supervised learning is used to train models on labeled datasets of emails (spam vs. not spam).
Algorithms learn to recognize patterns in:

- Email content (keywords, phrases)
- Sender behavior
- Metadata (e.g., frequency, links, attachments)

Real-world impact:
Spam filters in Gmail, Outlook, and other platforms now block **over 99% of unwanted emails**, improving productivity and security.

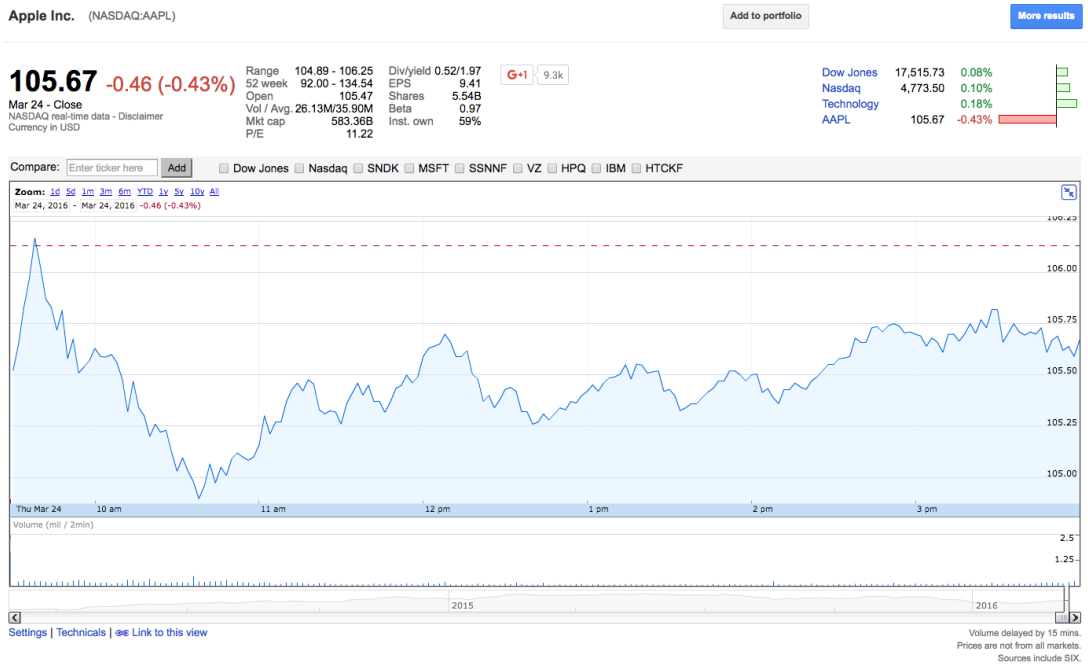


Figure source: http://www.resilientsystems.co.uk/

Machine Learning: Forecasting Stock Prices

Machine Learning models can be trained to **predict short-term movements** in stock prices, such as Apple's, using historical data and real-time market signals.

Real-world use:
Used by trading platforms, hedge funds, and financial analysts to support decision-making and automate trading strategies.

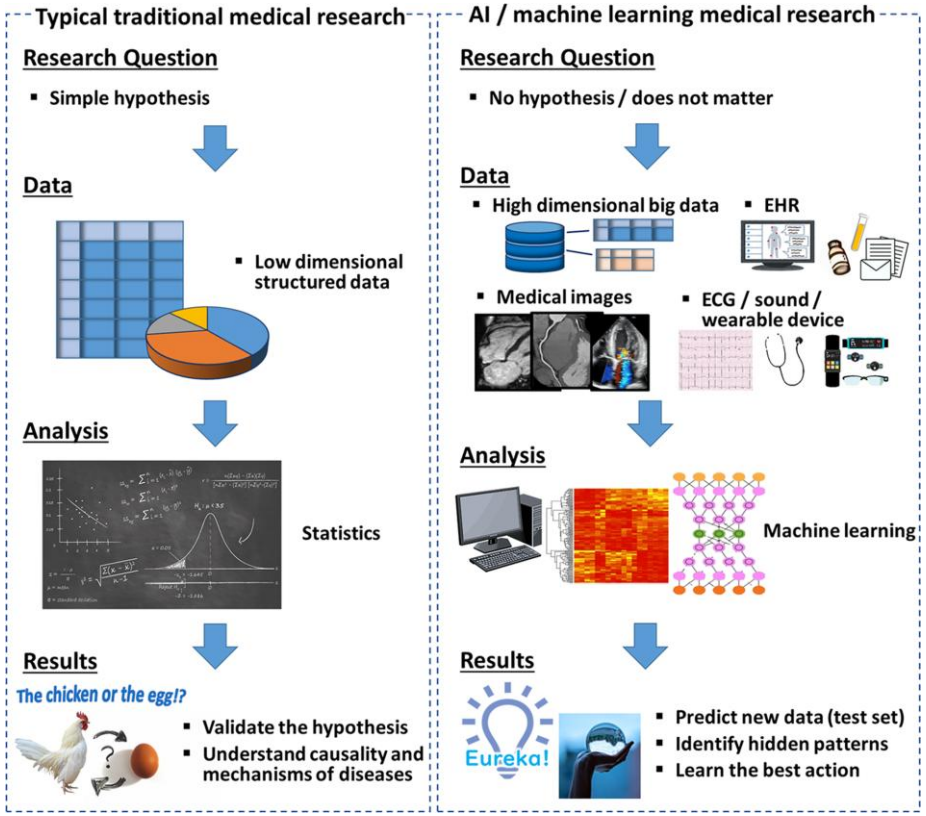


Machine Learning: Medical Diagnosis

- ML models are trained on large datasets containing **patient records**, **lab results**, **medical images**, and **clinical notes**.
- Algorithms learn to identify patterns and correlations that may not be obvious to human clinicians.

Real world applications

- Early detection of diseases like cancer, diabetes, and heart conditions.
- Support for doctors in making faster, more accurate decisions.
- Reduction in diagnostic errors and improved patient outcomes.



Machine Learning: Digit Recognition in Zipcode Forms

One of the classic applications of Machine Learning is **handwritten digit recognition**, used in postal systems to automatically read zip codes.

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Real-world impact:

- Speeds up mail sorting and delivery.
- Reduces manual errors and improves efficiency in logistics.

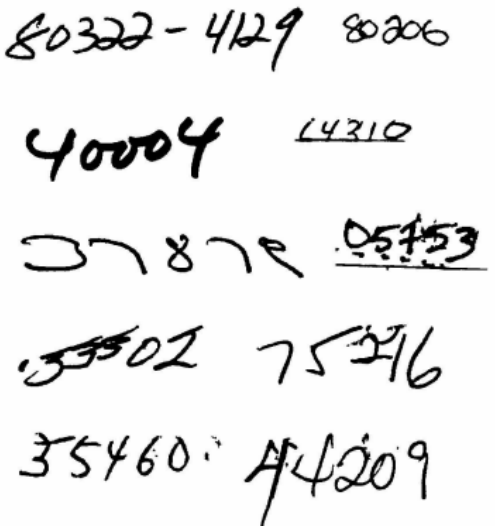


Figure source: [LeCun et al., 1989]

Machine Learning: Robotic Grasping

Machine Learning enables robots to learn how to **grasp objects**—like a mug—by interpreting sensory data and adapting to different shapes, weights, and positions.

How it works:

- Robots use **camera vision** and **depth sensors** to perceive the environment.
- ML models learn from trial and error how to position the gripper and apply the right force.
- Training involves **simulations** and **real-world feedback loops**.

Applications:

- Industrial automation (e.g., pick-and-place tasks)
- Service robots (e.g., assistive technologies)
- Warehousing and logistics

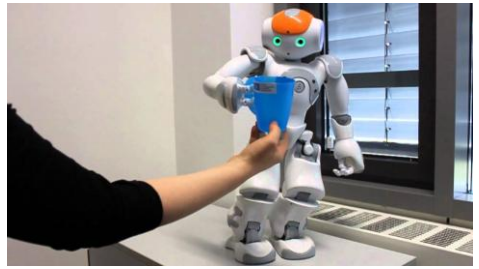


Figure source: http://www.informatik.uni-bremen.de/

Machine Learning: Drug and Molecule Design

Machine Learning is transforming pharmaceutical research by enabling the **design of new drugs and molecules** with specific biological or chemical properties.

How it works:

- ML models analyze vast datasets of molecular structures and known drug interactions.
- Algorithms learn to **predict molecular behavior**, such as binding affinity, toxicity, and solubility.
- Generative models can create entirely new molecular structures optimized for desired properties.

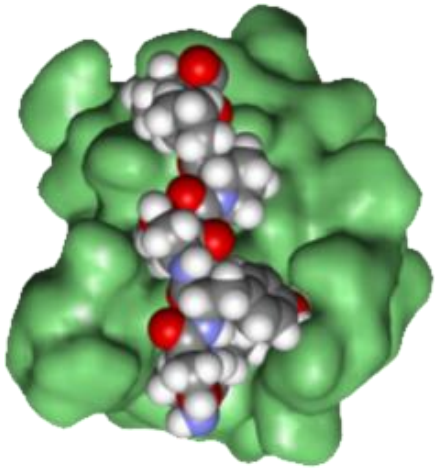


Figure source: http://pande.stanford.edu

Machine Learning: Machine Translation

How it works:

- ML models are trained on **parallel corpora**—large datasets of aligned text in multiple languages.
- Algorithms learn to map linguistic structures and meanings from one language to another.

Real-world impact:

- Used in tools like **Google Translate**, **DeepL**, and **Microsoft Translator**.
- Supports global communication, education, diplomacy, and commerce.



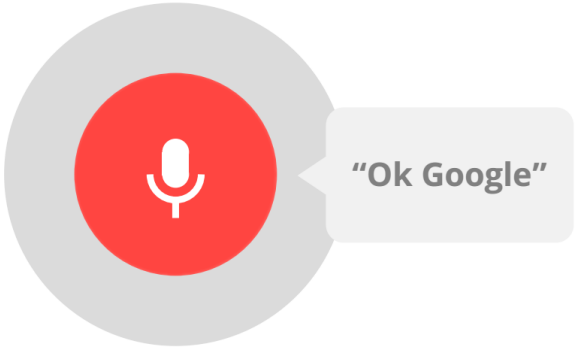
Machine Learning: Speech-to-Text

How it works:

- Audio input is processed to extract features like pitch, tone, and phonemes.
- ML models (especially deep learning) map these features to words and sentences.

Real-world applications:

- Virtual assistants (e.g., Siri, Alexa, Google Assistant)
- Automated meeting transcription (e.g., Zoom)



Machine Learning: Movie Recommendation

How it works:

- ML models analyze **user behavior**, such as viewing history, ratings, and search patterns.
- They use this data to predict what content a user is likely to enjoy.

Real-world impact:

- Increases user engagement and satisfaction.
- Drives personalized experiences and content discovery.
- Helps platforms optimize content delivery and production strategies.



Machine Learning: Image Captioning

How it works:

- **Convolutional Neural Networks (CNNs)** extract visual features from the image.
- **Recurrent Neural Networks (RNNs)** or **Transformers** generate a textual description based on those features.
- The model learns from large datasets of images paired with human-written captions (e.g., MS COCO).

Applications:

- Accessibility tools for visually impaired users
- Automated content tagging and indexing
- Enhancing search engines and digital assistants

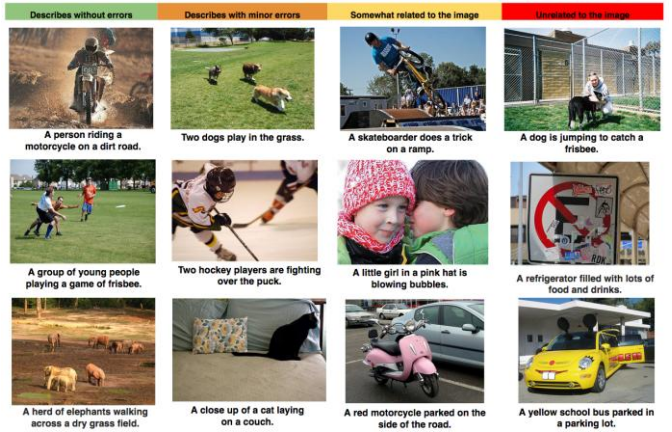


Figure source: Google Research

Machine Learning

AlphaGo Defeats the World's Top Go Player

In **March 2016**, Google DeepMind's **AlphaGo** made history by defeating **Lee Sedol**, one of the world's best Go players, in a five-game match.

- Marked a major leap in AI's ability to handle intuition, strategy, and creativity.
- Inspired further research in general-purpose learning systems.
- Sparked global interest in AI ethics, transparency, and human-AI collaboration.



Machine Learning: NLU and Reasoning

Natural Language Understanding (NLU) involves parsing and interpreting text to extract **meaning**, **intent**, and **context**.

Applications:

- Legal and medical document analysis
- Automated customer support
- Scientific research summarization
- Intelligent tutoring systems

GPT-4 Generating Template

Prompt: Andy harvests all the tomatoes from 18 plants that have 7 tomatoes each. If he dries half the tomatoes and turns a third of the remainder into marinara sauce, how many tomatoes are left?
Can you generate a template solution for the problem without numbers?

Using symbols, the template solution could look something like this:
- Let P be the number of plants and T be the number of tomatoes per plant. Then the total number of tomatoes harvested is P x T.
- Let D be the number of tomatoes that are dried. Then D = (P x T) / 2.
- Let M be the number of tomatoes that are turned into marinara sauce. Then M = ((P x T) - D) / 3.
- Let L be the number of tomatoes that are left. Then L = (P x T) - D - M.
...

[Figure from Bubek et al., 2023]

Main Machine Learning Paradigms

Supervised Learning

- **Goal**: Learn a mapping from inputs to known outputs.
- **Data**: Labeled examples (input-output pairs).
- **Applications**: Spam detection, image classification, medical diagnosis, stock price prediction.

Unsupervised Learning

- **Goal**: Discover hidden patterns or structures in data.
- **Data**: Unlabeled data (no predefined outputs).
- **Applications**: Customer segmentation, anomaly detection, frequent patterns, collaborative filtering

Reinforcement Learning

- **Goal**: Learn to make decisions by interacting with an environment.
- **Data**: Feedback in the form of rewards or penalties.
- **Applications**: Robotics, game playing (e.g., AlphaGo), autonomous vehicles.

Supervised Learning

Find a mapping between input/output variables, directly **from data observations**

- **Observed variables** $x_i \in X$
- **Prediction targets** $y_i \in Y$
- **Data set**, a collection of (x_i, y_i) pairs
- Build a **function** $f : X \rightarrow Y$
- Function f must be able to **predict** $\hat{y}$ **given some** $\hat{x}$

Supervised Learning

Different methods and problems according to:

- the kind of **input** space X
- the kind of **output** space Y
- the choices made for the **definition** of function f
  - Parametric models (e.g., linear models)
  - Non-parametric models (e.g., KNN)
- the choices made for the **computation** of function f

Supervised Learning

Classic input spaces

- X = numerical and/or attributes
- X = logic representations
- X = some structured space (tree, graph, etc.)

Supervised Learning

Classic output spaces

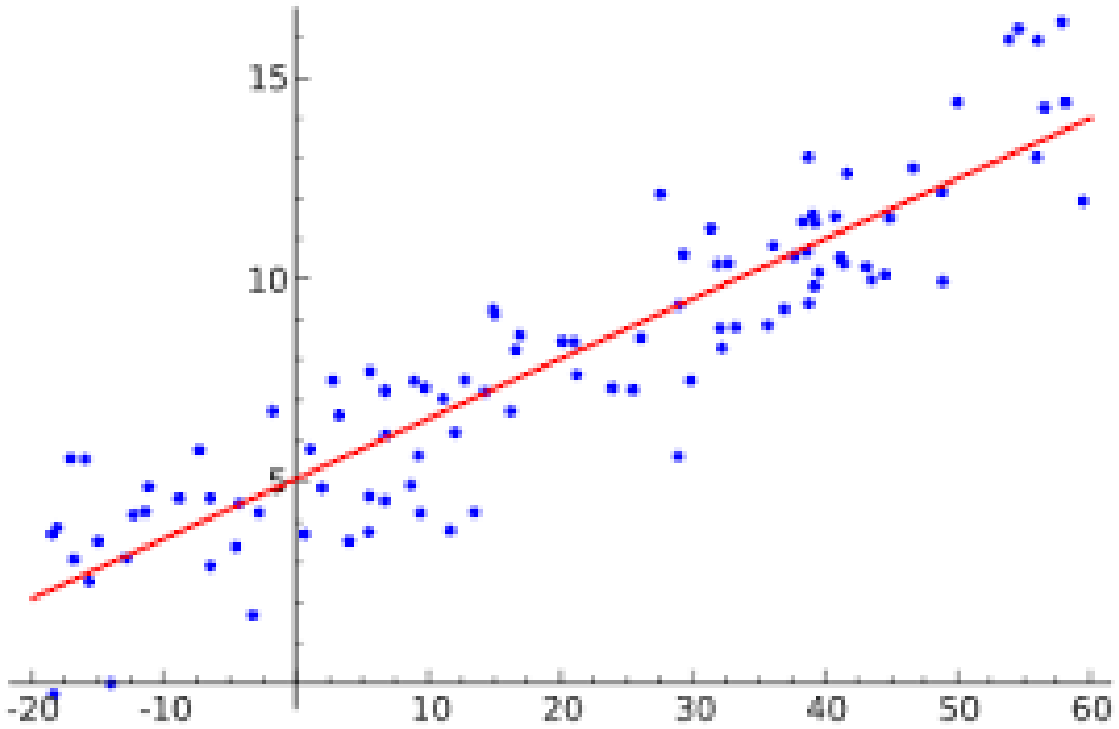
- $\mathrm{Y} = \{0, 1\}$ for binary classification
- $\mathrm{Y} = \{1, \ldots, M\}$ for multi-class classification
- $\mathrm{Y} = \mathbb{R}$ for regression
- $\mathrm{Y} =$ some structured space for structured output

Supervised Learning

A classic example: linear regression

- $\mathrm{X} = \mathbb{R}^d$
- $\mathrm{Y} = \mathbb{R}$
- $y = f(x) = \sum_{i=1}^{d} \alpha_i x_i$
- $\alpha = \arg\min \sum_{i=1}^{N} (\hat{y}_i - y_i)^2$

[Image from Wikipedia]



Measuring Performance

Given a **problem** and a machine learning **system** that solves it, it is crucial to **measure its performance**

Idea: **count the errors** of the system…
…Not so straightforward as one could expect!

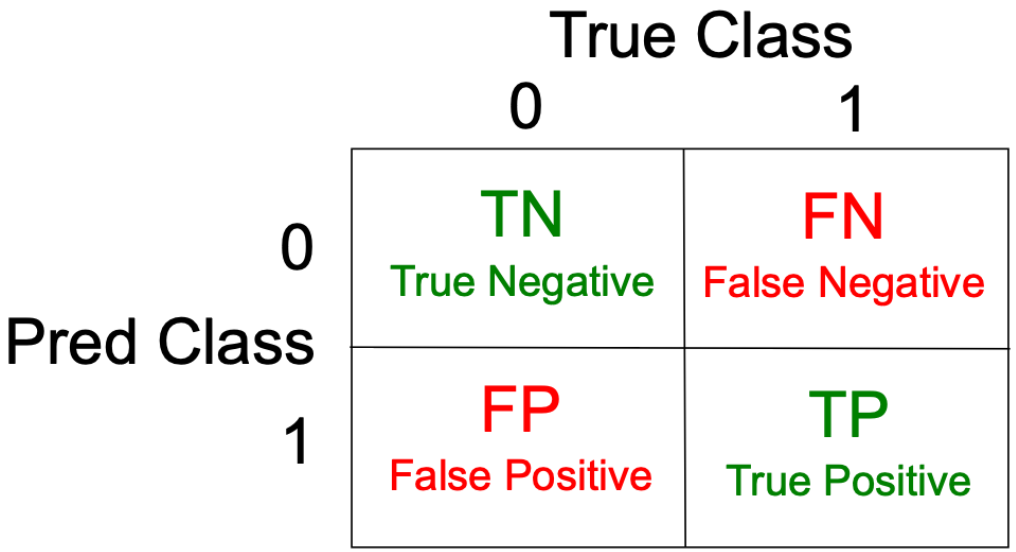
Measuring Performance

Example

We have to build a **diagnosis system** for a very **rare pathology**, based on machine learning

- Our system correctly classifies 99.9% of the cases
- Can we be happy with this classifier?

Confusion Matrix



$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$F_1 = \frac{2PR}{P+R}$$

Confusion Matrix

		True Class A	True Class B	True Class C
Pred Class	A	TP (A)	FP (A) FN (B)	FP (A) FN (C)
	B	FP (B) FN (A)	TP (B)	FP (B) FN (C)
	C	FP (C) FN (A)	FP (C) FN (B)	TP (C)

$$\text{Accuracy} = \frac{\sum_i TP_i}{\sum_i TP_i + FN_i} = \frac{\sum_i TP_i}{\sum_i TP_i + FP_i}$$

$$\text{Precision}^i = \frac{TP_i}{TP_i + FP_i}$$

$$\text{Recall}^i = \frac{TP_i}{TP_i + FN_i}$$

$$F_1^i = \frac{2P^i R^i}{P^i + R^i}$$

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What is best?

A)

Accuracy **91,7%**

Pred Class \ True Class	N	Y
N	5	0
Y	1	6

B)

Accuracy **71,4%**

Pred Class \ True Class	N	Y
N	400	350
Y	50	600

C)

Accuracy **71,4%**

Pred Class \ True Class	N	Y
N	400	50
Y	350	600

D)

Accuracy **85,7%**

Pred Class \ True Class	N	Y
N	600	100
Y	0	0

Regression Metrics

For regression problems

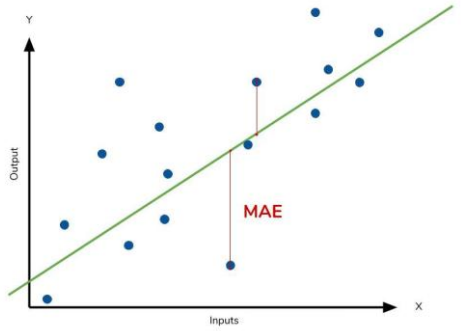
- Mean squared error (MSE)
- Root mean squared error (RMSE)

$$\text{MSE} = \sum_{i=1}^{N} \frac{(\hat{y}_i - y_i)^2}{N}$$

$$\text{RMSE} = \sqrt{MSE}$$

$$\text{MAE o MAD} = \frac{1}{N} \sum_{i=1}^{N} |\hat{y}_i - y_i|$$

- Mean absolute deviation (MAD)
- Mean absolute error (MAE)



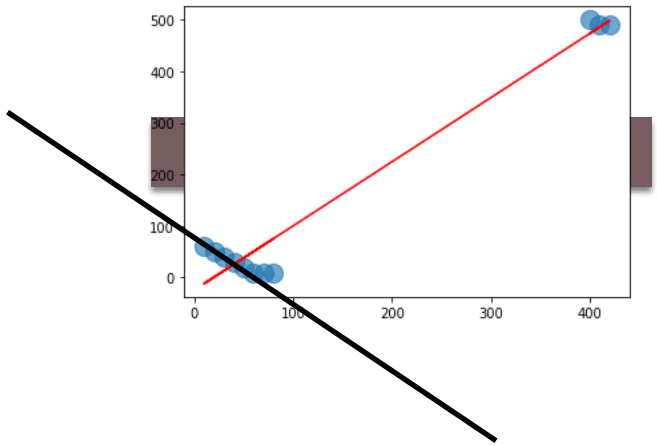
- R^2 Explained Variance

$$R^2 = 1 - \frac{\text{Area of green squares}}{\text{Area of pink squares}}$$

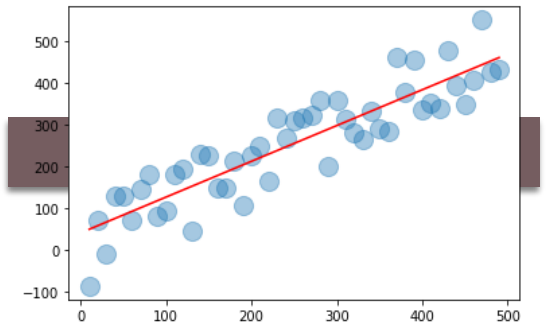
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What is best?

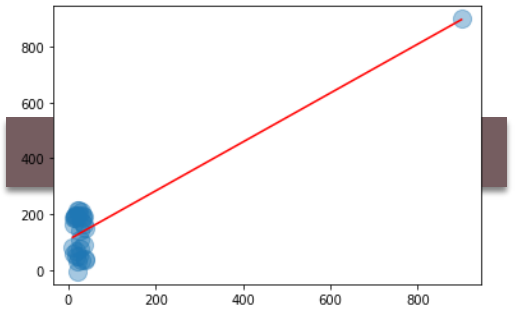
MAE = 33.5
R2 = 0.96



MAE = 52.1
R2 = 0.80



MAE = 61.96
R2 = 0.82



Model Selection

The performance of any machine learning system **strongly depends** on a set of **hyper-parameters**, whose values have to be chosen during the training phase

The **procedure** through which we choose the best hyper-parameters for a machine learning system is called **model selection**

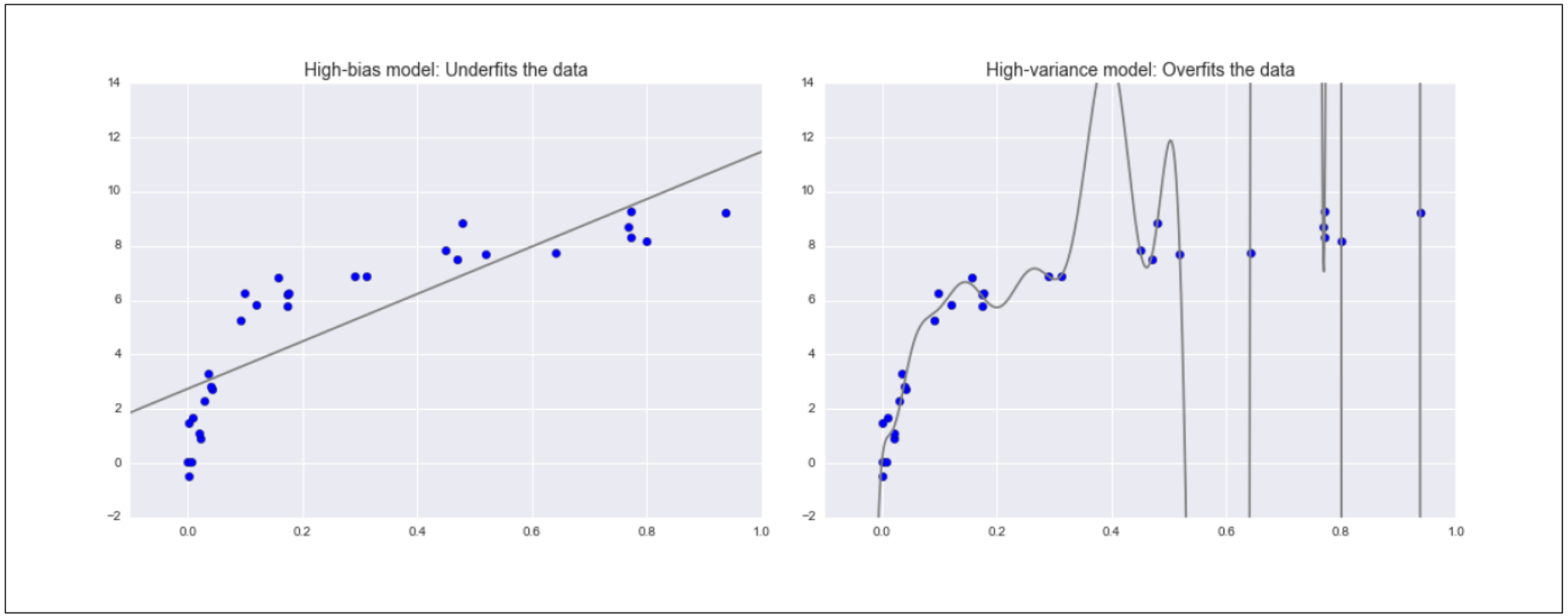
Model Selection

How to build and choose a model?
Split the data into:

- **Training set** (to learn the objective function)
- **Validation set** (to select the best model)
- **Test set** (to measure performance on unseen data)

If possible, maintain **proportions** among classes

Bias-Variance Tradeoff - Underfitting Vs. Overfittnig

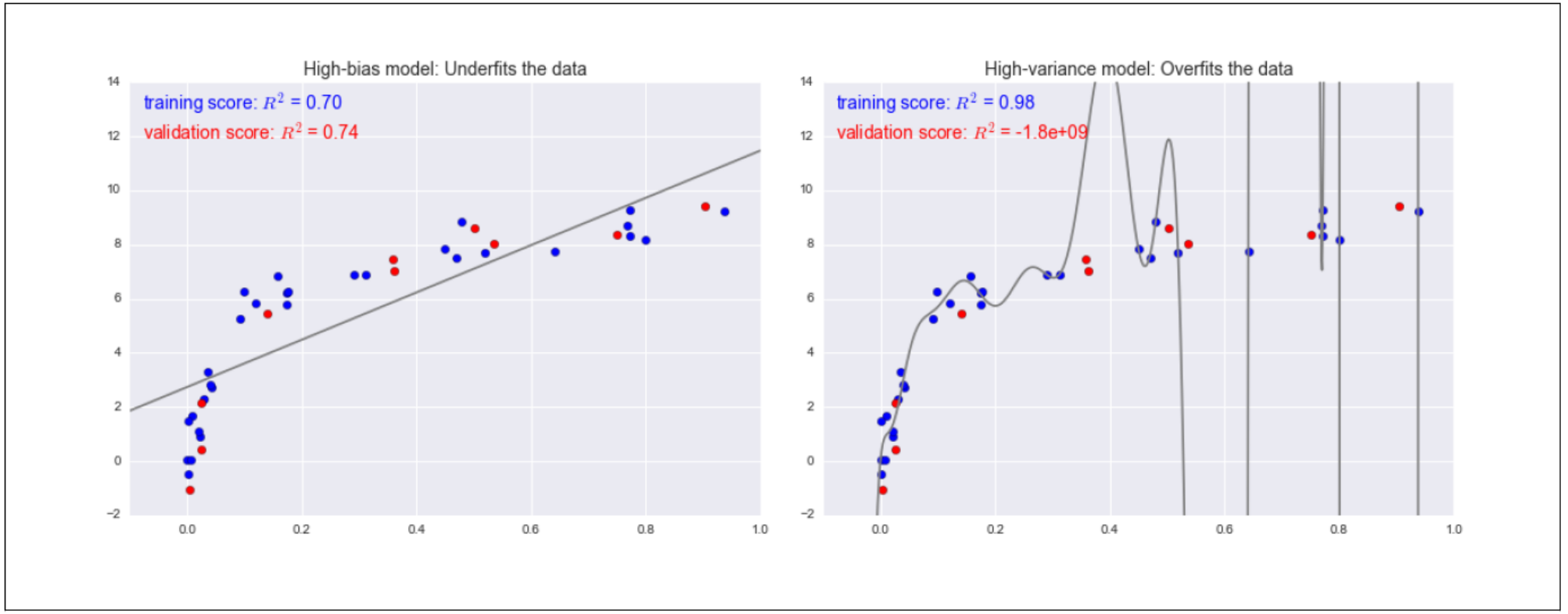


Model has **not enough flexibiilty**
It **underfits** the data
It has **high bias**

Model has **too much flexibiilty**
(that can accout also for noise)
It **overfits** the data
It has **high variance**

What happens if new data come around?

What are the performance with new data?



For high-bias models, the performance of the model on the validation set is similar to the performance on the training set

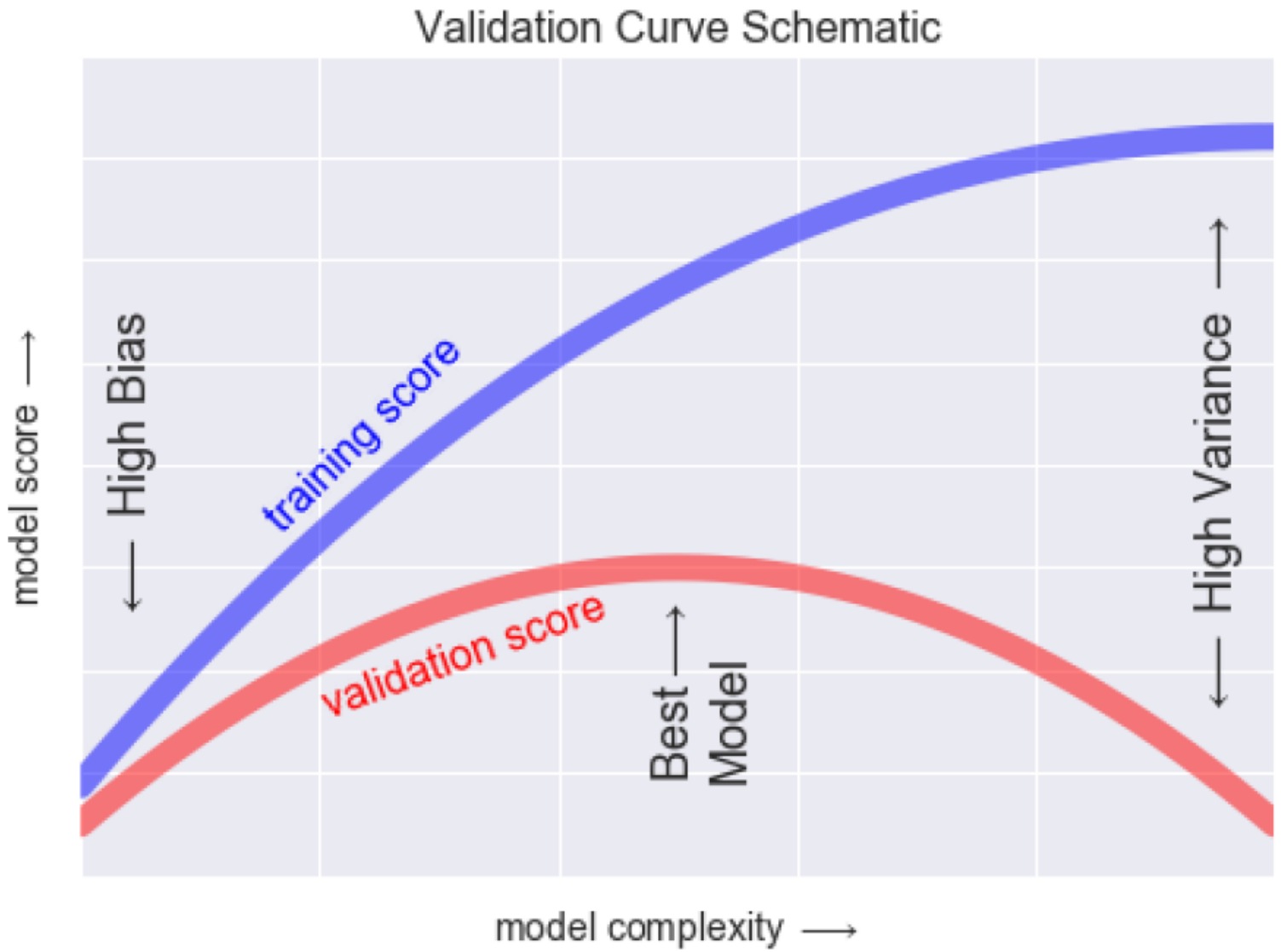
For high-variance models, the performance of the model on the validation set is far worse than the performance on the training set

Training, Validation (Testing) Set

- Therefore, it is important to evaluate the model on data different that the one used for training.

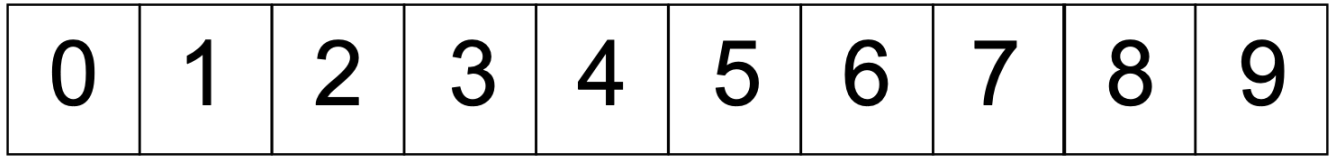
- Part of the data is used to train (fit) the model
- Part of the data is used for evaluation

Bias-Variance Error

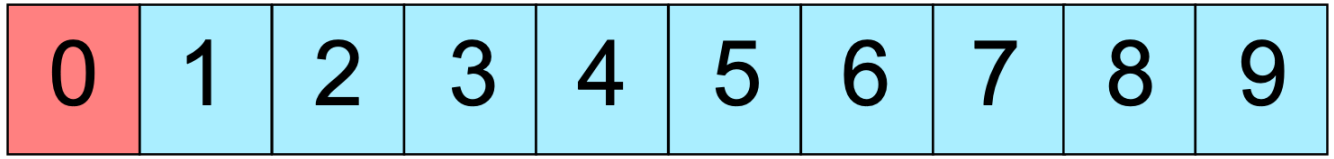


K-Fold Cross Validation

Split the data into K sets, named **folds**

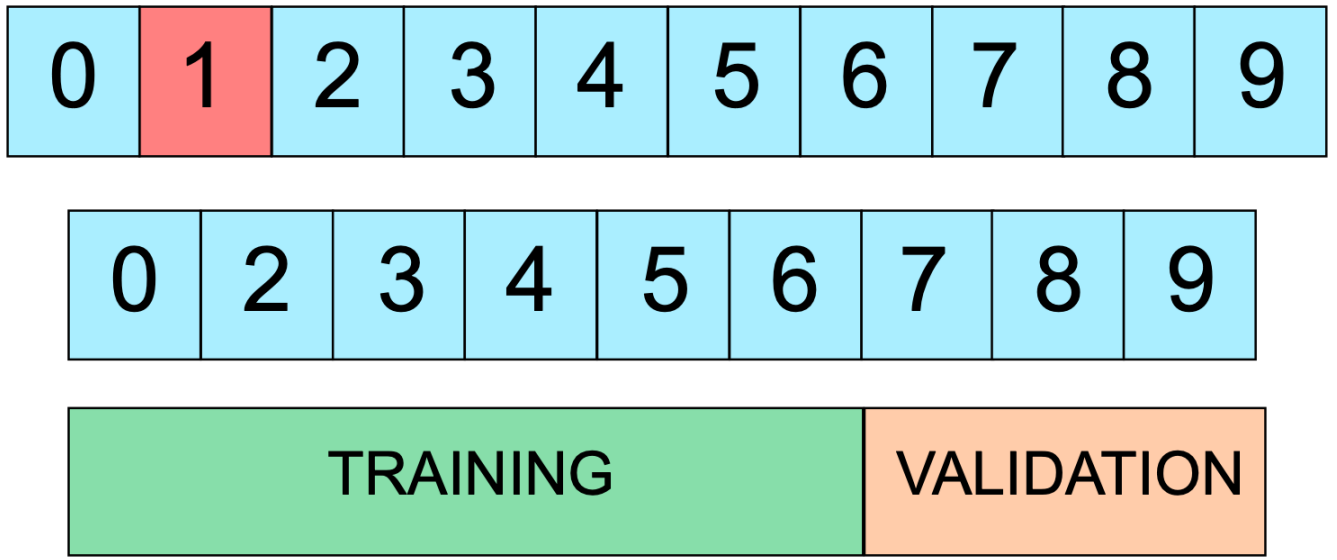


In turn, each fold will act as **test set**, whereas the remaining will form the **training and validation sets**



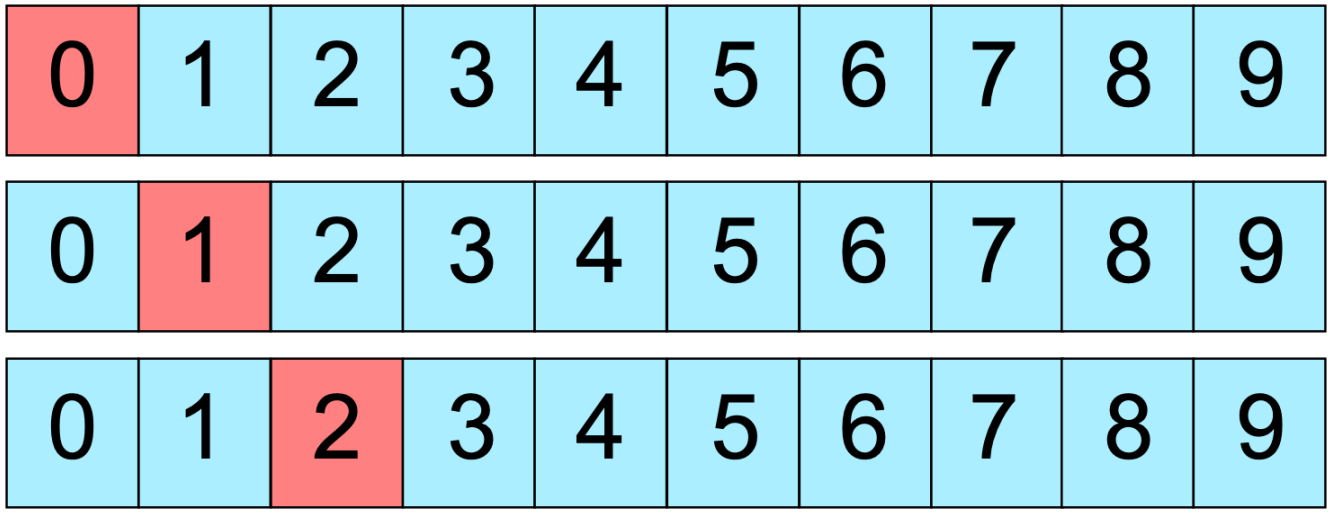
K-Fold Cross Validation

Training and validation sets are used to select the model. Test set is used to compute the performance



K-Fold Cross Validation

The procedure is repeated K times, for all the folds



Macro- vs. Micro-Average

Statistics computed **over the K folds** are reported.

Suppose to have the following results from a 3-fold cv:

- TP=5, TN=10, FP=2, FN=3
- TP=3, TN=12, FP=4, FN=1
- TP=8, TN=11, FP=0, FN=1

Two possibilities:

- compute one metric per fold, then average (**macro**)
- sum all TP, FP, FN, then compute one metric (**micro**)

Macro- vs. Micro-Average

Example — Macro-Precision

- TP=5, TN=10, FP=2, FN=3
- TP=3, TN=12, FP=4, FN=1
- TP=8, TN=11, FP=0, FN=1

$P_1 = 5/7 = 0.714$

$P_2 = 3/7 = 0.286$

$P_3 = 8/8 = 1.000$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$P_{Macro} = (0.714 + 0.286 + 1.000) / 3 = \mathbf{0.667}$

Macro- vs. Micro-Average

Example — Micro-Precision

- TP=5, TN=10, FP=2, FN=3
- TP=3, TN=12, FP=4, FN=1
- TP=8, TN=11, FP=0, FN=1

P_{micro} = 16/22 = **0.727**

Overfitting

A machine learning system is usually trained to **minimize some cost (loss) function**, thus it should not be surprising that it is capable to have a low error on the examples seen during training!

Main drawback

- One could **perfectly** learn the training set, while being completely **unable to generalize** across different examples!
- This is called **overfitting**

Overfitting

How to reduce overfitting?

- Validate model on data **not seen** during training
- Include a **regularization** term (more on this later)

Data Preprocessing

A crucial step in machine learning is to **prepare data** so that the **useful** information can be easily exploited

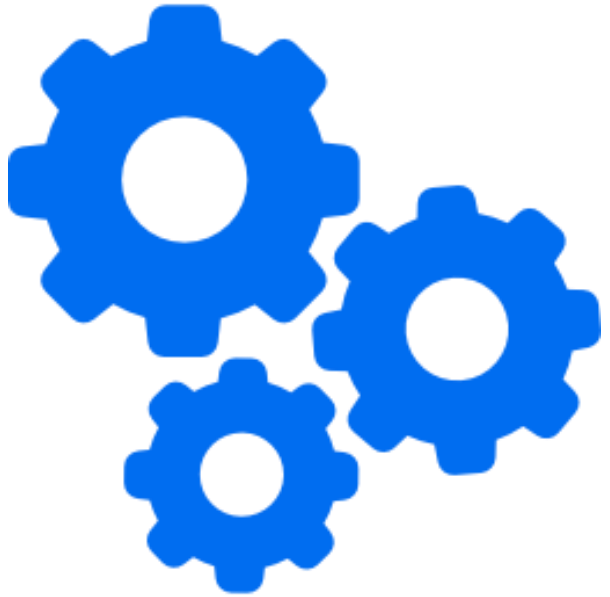
Data have to be formatted according to:

- Model/algorithm
- Software tool

Data Preprocessing

Main issues

- Integration
- Cleaning
- Normalization
- Aggregation
- Conversion
- Discretization
- Dimensionality reduction



Data Preprocessing

Integration

- Combine **several** different data sources
- Convert one **format** into another
- Select or **sample** data, if necessary
- Handle **units** of measurement
- Remove **duplicates**



[Image from naniinfotech.com]

Data Preprocessing

Cleaning (or cleansing)

- Handle **heterogeneous** values
- Handle **missing** data
  - **Ignore**: some models/algorithms can handle them
  - **Remove**: drop either examples or attribute
  - **Substitute**: with average/median/mode
- Check for **inconsistent** data (e.g. city/zipcode)
- Check for **outliers** (maybe due to mistakes)



Data Preprocessing

Normalization

- Some algorithms need the **input data** to be in a specific range of values (e.g. [0,1] or [-1,1])
- In general, it is a good practice to re-scale attribute values so that they are in **similar ranges**
- **Avoid** the fact that some attribute is **numerically too small/large** with respect to the others

Data Preprocessing

Min-Max normalization $$\hat{v} = \frac{v - v^{\min}}{v^{\max} - v^{\min}}$$

Mean-Std normalization (a.k.a. standardization) $$\hat{v} = \frac{v - \mu}{\sigma}$$

Data Preprocessing

Aggregation

- In many cases one needs to **group** some data according to some attribute value
- Typical in time-series, where one can choose the data **granularity** to be used for data representation

Data Preprocessing

Conversion

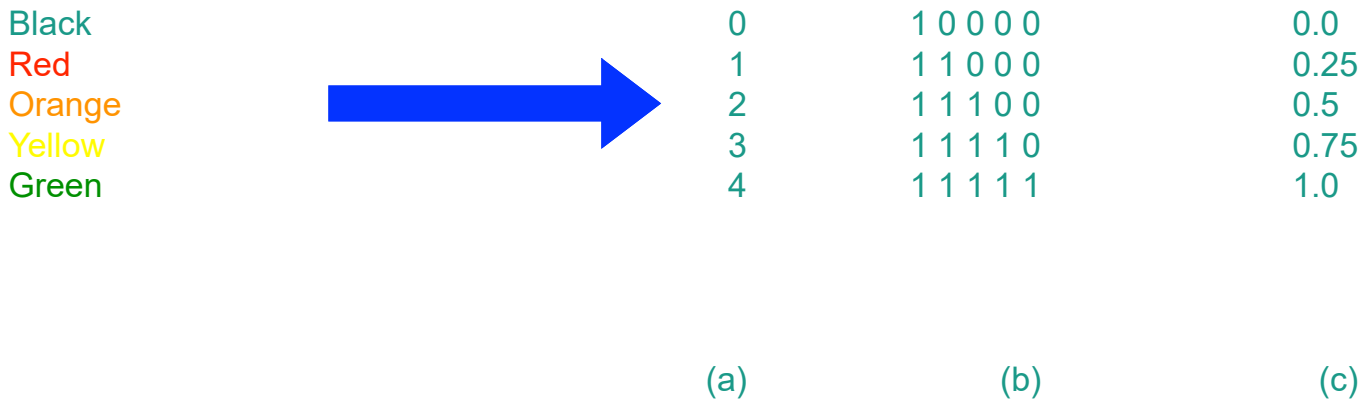
- Many algorithms can only handle **numeric** values
- Some attributes are **discrete** or **categorical**
- In the first case convert to **numeric** (ranked) values
- In the second case exploit **one-hot encoding**

Data Preprocessing

Ordinal to numeric conversion

- Preserve the **ordering** between discrete values

Example (Google traffic colormap):



Data Preprocessing

One-hot encoding

- Transform a **categorical** attribute with N possible values into a set of N **binary flags** (0/1)
- Example:

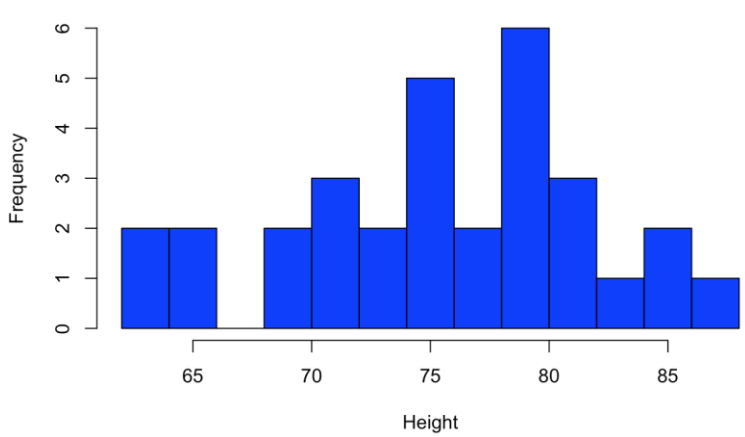


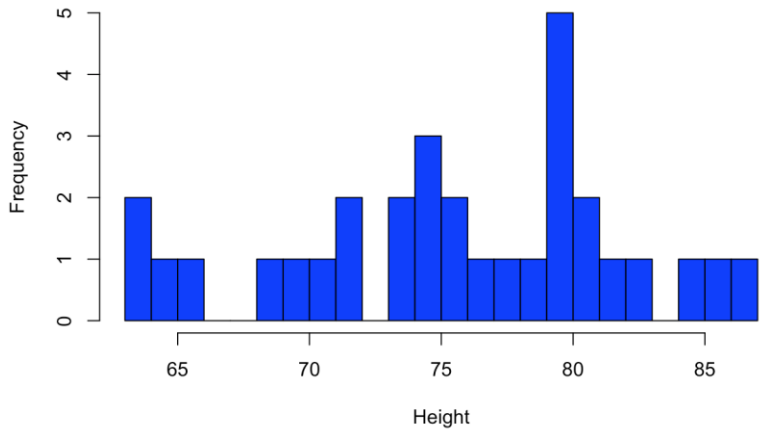
Note: all values are equally distant one from the other

Data Preprocessing

Discretization

- Some algorithms **cannot** handle numerical features
- Necessary to perform **quantization**
- Choose number of **bins** in the histogram





Data Preprocessing

Discretization

- **Equal-width binning**: partition the range into K bins, each having the same width
- **Equal-frequency binning**: partition the range into K bins, each having the same number of instances

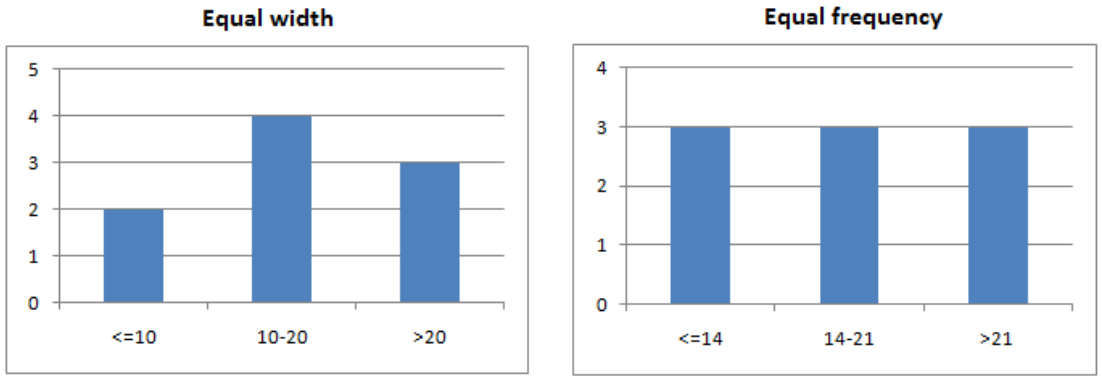
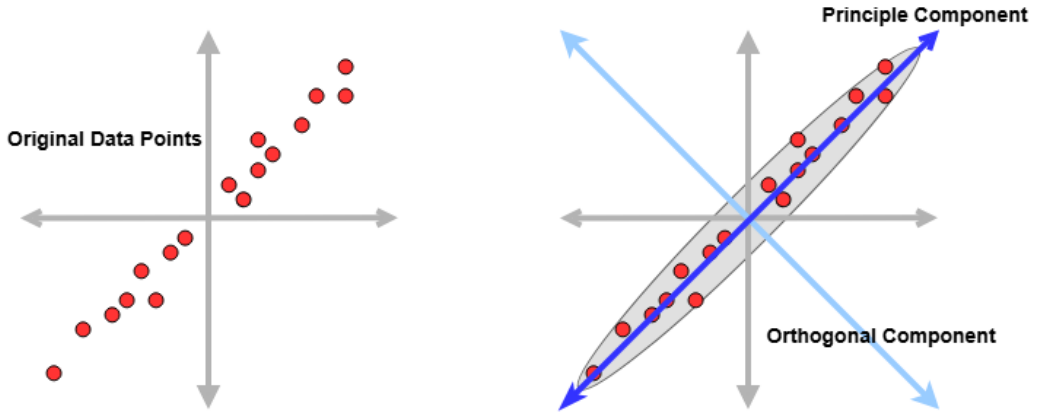


Image from http://www.saedsayad.com/

Data Preprocessing

- Dimensionality Reduction
  - PCA
  - Autoencoder
  - …

- **Principal Component Analysis (PCA)** is a technique for mapping a set of examples, represented by many features, to a lower-dimensional space, where the main data characteristics are preserved. It is particularly useful for exploratory data analysis, for example to visualize data with many features



Example on Time-Series

Parameters to control when forecasting time-series?



DATA AGGREGATION?